

Technology #10: Autonomous Catheter Navigation with Human-in-the-Loop

Technology Domain: Catheter Mechanics + Hydraulic Actuation + Tissue Interaction

Mechanism: Steerable CSF shunt catheter with hydraulic actuation for autonomous navigation during placement. R1 fix: original P-22 assumed fully autonomous navigation without addressing buckling failure modes; R1 redesign adds buckling analysis, tissue damage safety, and human-in-the-loop fallback.

Maturity: ENGINEERING_DEFINITION

Current State: No physical prototype. Computational model and engineering definition exist.

Key Risk: Buckling threshold is below typical insertion forces (catheter cannot navigate without buckling)

Next Experiment: Catheter mechanical specimens (various EI)

Buyer Receives: Autonomous navigation concept + R1 correction (buckling + tissue safety + human-in-the-loop); Beam theory + buckling + hydraulic governing equations; Bench test protocols (buckling, navigation accuracy, tissue damage, biocompatibility); External precedent catalog (steerable catheter literature); Critical UNKNOWN disclosure (tissue damage threshold, achievable accuracy); This engineering dossier with failure analysis + build plan

Buyer Must Build: Production catheter design (multi-lumen, actuator integrated); Production navigation controller (software V&V; per IEC 62304); Manufacturing process for steerable shunt catheter; Regulatory submission (autonomous navigation adds complexity; likely Class III); Clinical validation (placement accuracy endpoint); Tissue damage threshold establishment (animal studies)

Transfer Posture: Sponsored Validation

Target Buyer: Surgical robotics company; neuroshunt manufacturer

CONFIDENTIAL. See full engineering dossier for details.